

UCL N2O Report

Methodology

Codebase Implementation

For the purposes of this analysis, I developed: [respiratory-complexity-analysis](#). You can use this codebase to play around and test the influence of different parameters on the final LZC results. Its main page / readme file provides a description of what it does and how to use it.

Phase 0: Received raw files

The received raw files provided in 'N2O-Study2-BIOPAC' folder (renamed to 'data_raw' in the repository) had their line noise removed and were bandpass filtered 0.05-5Hz.

Phase 1: Signal Preprocessing

1. The signal was filtered using the 2nd-order 0.05-3 Hz bandpass Butterworth filter.
 - 'khodadad2018' method (inspired by [Khodadad et al. \(2018\)](#)) of [NeuroKit2](#) (a Python library for biosignal processing) was used to filter the signal.
 - It removes the baseline drift and preserves breathing rates in the range of 3-180 breaths per minute.
2. The filtered signal was downsampled from 2000Hz to 200Hz.
 - A visual analysis showed that 25Hz, or maybe even lower, also preserves the signal structure. The 200Hz results were compared with 400Hz and 25Hz results.
 - `complexity_sensitivity_frequency.py` and `plot_different_sampling_rates.py` allow exploration of these differences in more detail.
3. The downsampled signal was segmented into continuous pre-inhalation and during-inhalation segments, avoiding including any interruptions in the middle.
 - Segmentation was done using .txt files from the 'N2O-Study2-BIOPAC'.
 - A 'User Type 9' timestamps were used as indicators of starting and ending inhalation, and 'Append' timestamps were interpreted as breaks in the recordings which continuous segments should not include.

4. The segmented signal was then split further into 120s long epochs.
 - The 120s window was determined by implementing a complexity sensitivity analysis which looked at what point does the complexity of the signal stop changing by more than 2% between windows.
 - Sensitivity analysis was implemented in `complexity_sensitivity_window.py`.
5. Bad epoch detection was not implemented (WIP).
 - Due to no standardized method of epoch rejection for respiratory data (as in for example with EEG epoch rejection in the MNE Python library), I attempted to implement a manual one but concluded that it would be better to first consult with you with regards to what do you think it should reject.

Phase 2: Lempel-Ziv Complexity Calculation & Analysis

1. The epochs were binarized around the median value. The LZC value was calculated for each epoch, and averaged across subject-condition pairs and conditions (pre-inhalation and n2o-inhalation / air-inhalation).
2. T-tests were done to compare mean LZC results across conditions.
3. The participant data (gender, subjective questionnaires, etc.) was used to fit a Robust Linear Model (RLM). Compared to a traditional regression analysis, this model puts less emphasis on the outliers. Three different RLM models were explored, fit to predict LZC:
 - across 3 conditions separately (pre, air, n2o).
 - across 2 phases (pre, during)
 - across all data, using the drug (n2o, air) and phase (pre, during) as predictors.

Remarks

- Results were compared to standardized results (standardized on the file-level, segment-level, and epoch-level) and this showed no difference in the results. This is likely due to filtering out the baseline drift and the nature of the median binarization for LZC calculation.
- More technical code details and tools used can be found in the repository's [readme file](#).

Results

All results presented below were obtained with the following parameters:

- filtering method: **khodadad2018**
- epoch length: **120s**
- downsampled to: **400Hz, 200Hz, 25Hz**

Mean LZC across conditions

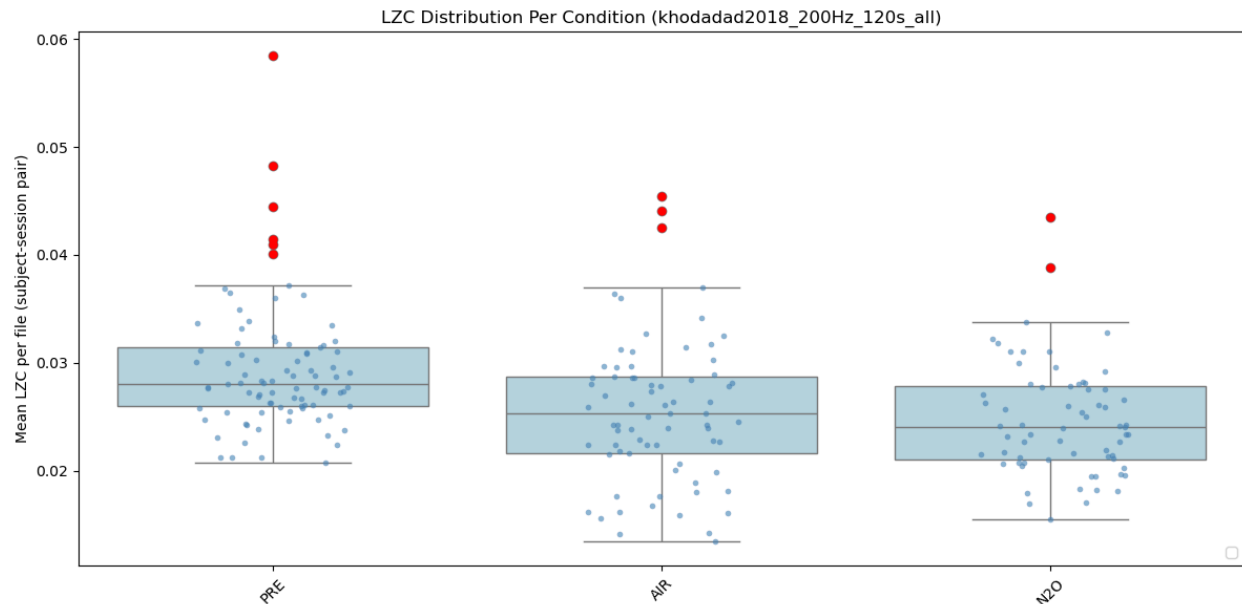
	Conditions	400Hz	200Hz	25Hz
Mean LZC	PRE (88 subject-sessions)	0.0158 ± 0.0032	0.0294 ± 0.0060	0.1721 ± 0.0320
	AIR (73 subject-sessions)	0.0137 ± 0.0037	0.0255 ± 0.0068	0.1511 ± 0.0366
	N2O (67 subject-sessions)	0.0132 ± 0.0028	0.0247 ± 0.0052	0.1480 ± 0.0284

T-test across conditions

	Conditions diff	400Hz	200Hz	25Hz
p-value	AIR vs PRE	0.0001945	0.0001874	0.000187
	N2O vs PRE	0.0000004	0.0000005	0.0000019
	N2O vs AIR	0.382926	0.4112947	0.5720515

RLM analysis & significant variables

Displayed RLM was fit using the phase (pre and during inhalation) and drug (medical air and n2o) as predicting variables, across all files.

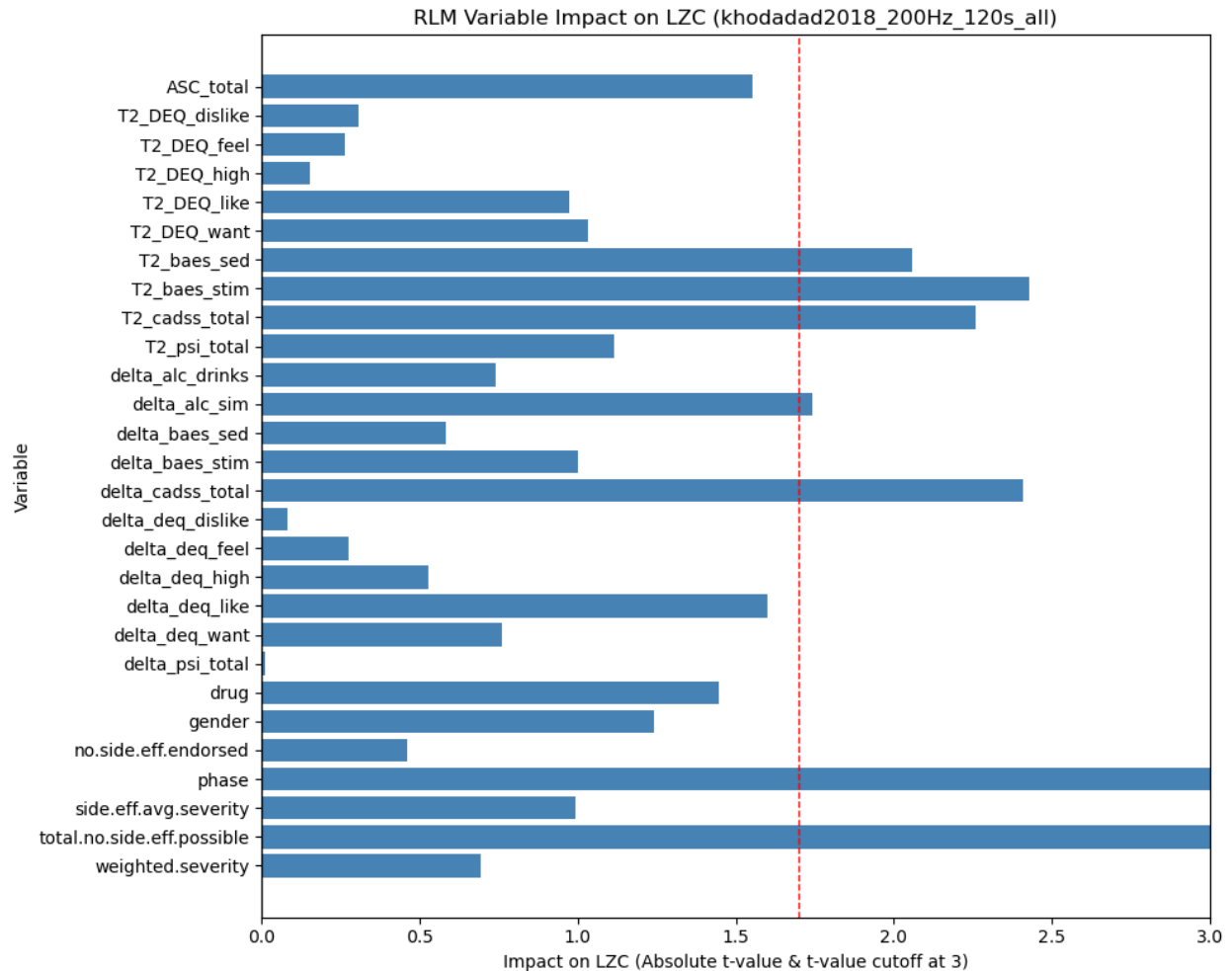


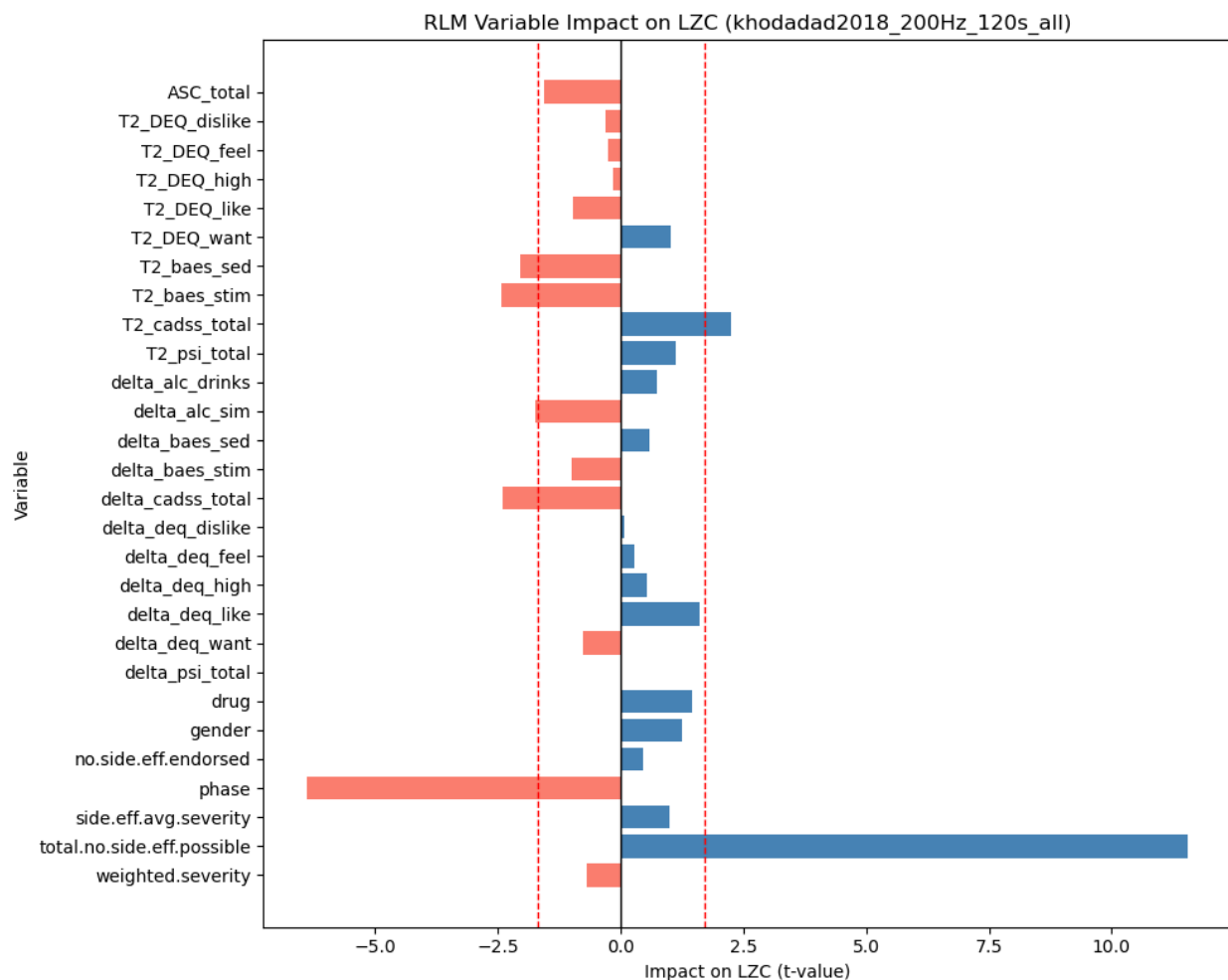
The outliers marked in red are the following:

P006-1, P026-1, P026-2, P028-2, P037-1, P043-1, P063-1, P067-1, P067-2, P073-1, P076-2.

The barplots below show same significant RLM variables in 2 different plots:

1. with absolute t-values cutoff at 3
2. with positive and negative t-values





The results below are selected based on their t-value (≥ 1.7) significance, with the slightly stricter p-value (≤ 0.05) displayed.

	Significant RLM vars	400Hz	200Hz	25Hz
p-value	drug	(t not t significant)	(t not significant)	0.073268
	phase	0	0	0
	T2_baes_stim	0.015847	0.015227	0.020771
	T2_baes_sed	0.039807	0.039736	0.053648
	T2_cadss_total	0.02934	0.023997	0.013265
	delta_deq_like	(t not significant)	(t not significant)	0.079324
	delta_cadss_total	0.019611	0.015963	0.007921
	delta_alc_sim	0.086708	0.081239	0.071013
	total.no.side.eff.possible	0	0	0

Summary

LZC of respiratory data appears to drop during inhalation, both with medical air and N2O, with N2O dropping slightly lower.

However, t-test showed no significant difference in complexity between the N2O group and control group. The only significant difference was observed between pre-inhalation and both air-inhalation and N2O-inhalation.

Robust Linear Model analysis also did not highlight the 'drug' as a relevant variable, only the 'phase'. Not even when the RLM was exclusively fitted only onto the during-inhalation data, omitting pre-inhalation data.

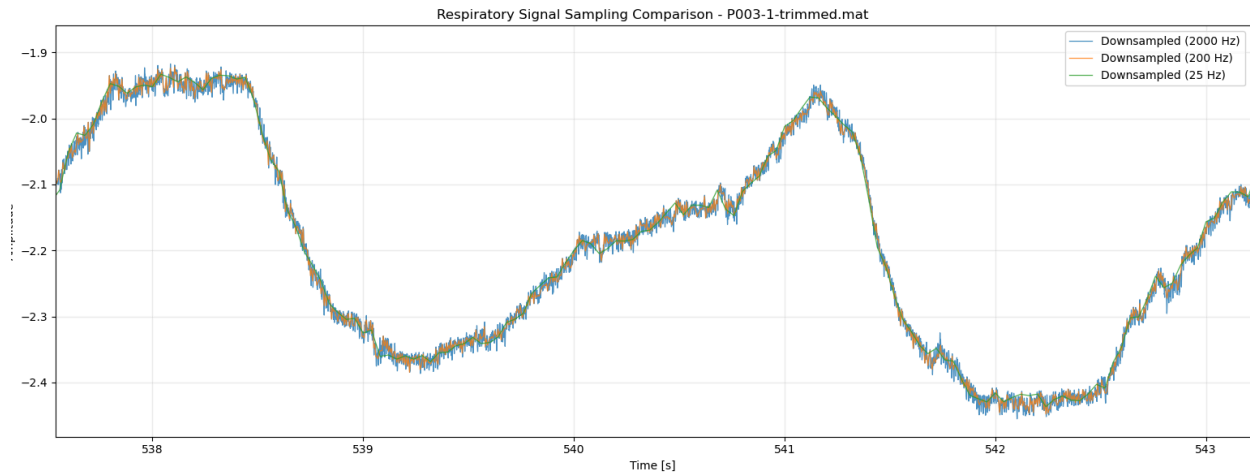
This overall points at complexity decreasing during inhalation, irrelevant of the gas being inhaled. It is possible that I messed up something in the calculations and will be happy to discuss this in more detail with you to see if I missed anything.

The variables which RLM does highlight as relevant are:

- drug (*interestingly only becomes significant when downsampled to 25Hz*)
- phase (pre inhalation VS during inhalation)
- Stimulation subscale of BAES at peri-inhalation (T2_baes_stim)
- Sedation subscale of BAES at peri-inhalation (T2_baes_sed)
- Total score on the CADSS at peri-inhalation (T2_cadss_total)
- Delta of 'Like' subscale of DEQ (delta_deq_like)
- Delta of total score on the CADSS (delta_cadss_total)
- delta_alc_sim (?)
- Total number of possible side effects participants were asked about (total.no.side.eff.possible)

Appendix A: Preprocessing Analyses

Downsampling the signal to 25Hz maintained the signal's structure overall, as can be maintained by a smooth green line passing through the noisy lines.



The complexity sensitivity analysis for window sizes stabilized at 120s, as is observable by the green dot registering a change of less than 2%.

